

Table 49. Output AC characteristics (all I/Os except PC13)

The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT_a refers to any TT I/O with _a option. TT_xx refers to any TT I/O and FT_xx refers to any FT I/O.

The I/O speed is configured using the OSPEEDRy[1:0] bits. Refer to the product reference manual for a description of GPIO port configuration register

Specified by design - Not tested in production.

The minimum and maximum values are specified for a junction temperature $T_J = 125^{\circ}\text{C}$.

Speed	Symbol	Parameter	conditions	Min	Max	Unit
00	Fmax ⁽¹⁾	Maximum frequency	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	4	MHz
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	4	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	4	
	tr/tf ⁽²⁾	Output high to low level fall time and output low to high level rise time	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	51	ns
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	46	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	40	
01	Fmax ⁽¹⁾	Maximum frequency	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	12	MHz
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	12	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	12	
	tr/tf ⁽²⁾	Output high to low level fall time and output low to high level rise time	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	19	ns
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	17	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	14	
10	Fmax ⁽¹⁾⁽³⁾	Maximum frequency	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	45	MHz
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	50	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	55	
	tr/tf ⁽²⁾⁽³⁾	Output high to low level fall time and output low to high level rise time	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	7	ns
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	6	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	4	
11	Fmax ⁽¹⁾⁽³⁾	Maximum frequency	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	90	MHz
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	96	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	110	
	tr/tf ⁽²⁾⁽³⁾	Output high to low level fall time and output low to high level rise time	C = 50 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	5	ns
			C = 30 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	4	
			C = 10 pF, 2.7 V ≤ V _{DD} ≤ 3.6 V	-	3	

1. The maximum frequency is defined with the following conditions: $(tr+tf) \leq 2/3 T$

2. The fall and rise times are defined between 90% and 10% and between 10% and 90% of the output waveform, respectively.

3. Compensation system enabled.